

Ultra Low Resource LoRA Fine-Tuning for Mitigating LLM Bias in Medical Question and Answering

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Abstract—The following report details our experiments on using ultra low resource LoRA Fine-Tuning for Mitigating LLM Bias in Medical Question and Answering. One open source medical LLM model and two generalize open source LLM models were used to determine if models could be fine-tuned with a small subset of examples to deliver more accurate and less biased responses to medical question and answering tasks. This methodology was tested on three prompting strategies: Zero-Shot, Few-Shot, and Chain-of-Thought prompting. We were able to find significant increases in cosine similarity across post fine-tuned models that used zero-shot prompting and smaller performance improvements across other prompting types. Overall, our results show that even with using a small number of samples for fine-tuning, some bias can be resolved and model correctness can be improved with our provided method.

I. INTRODUCTION

Large Language Models (LLMs) have demonstrated significant potential in the healthcare domain, with applications ranging from summarizing clinical notes to providing preliminary patient consultations [1]. These health applications give consultation to patients for a reduced cost [2]. Their value lies in making medical consultations more convenient and reducing healthcare costs.

However, these medical LLM models exhibit bias. In the context of healthcare, bias means the LLM outputting unfair responses to patients of a certain sociodemographic or specific attributes like history of a disease [3]. It has been challenging to examine the biases in LLMs in medical practice [3]. Different types of biases such as data centric bias, inference bias, and model parametric bias have been researched with respect to medical LLMs [3]. Data centric bias refers to bias that comes from the training data [4]. Inference bias refers to bias caused by different prompting strategies [5]. Model parametric bias refers to bias from different model architectures and parameter size [6]. Furthermore, different strategies to mitigate the bias have been researched, such as data model design [3]. Data model design refers to using a variety of datasets to mitigate bias [4]. Inference procedure refers to changing the inputs and outputs to reduce bias [7]. To use medical LLMs in medical practice, they must produce outputs that are both reliable and accurate. Furthermore, in the context of healthcare, it isn't enough for healthcare applications to perform well on paper. Healthcare applications need to sustain clinical significance to help patients in the long run [3].

Because of the biases in the medical LLMs, medical LLMs currently have low accuracy [2]. They can also give misleading information to their patients. One example of this is where the medical LLM gave advice to a patient with fever and stomachache to rest. In reality, the medical LLM overlooked some changes like appendicitis [2]. Therefore it is important that the use of these medical LLMs is accompanied by a doctor's oversight [2].

Fine tuning involves further training the model on a specific task dataset, which helps it specialize in a particular area. By further training the model on a specific dataset, fine tuning can increase accuracy, which can reduce bias and prevent hallucinations. Another benefit to fine tuning is that it is cost effective, fine tuning can be cheaper than training a new model completely from scratch [8].

Next, let us look at a study that shows the benefits of fine tuning. In a study by MBIAS: Mitigating Bias in Large Language Models While Retaining Context, they created MBIAS, which is an instruction-fine tuned model built on top of mistral-7B. They compared instruction-finetuned LLMs of mistral-7B to several baseline models, such as a prompt tuned and vanilla versions of mistral-7B and llama2-7B. According to the study, "MBIAS, shows a significant reduction in bias (9.49%) and toxicity (8.71%), while achieving the highest score in knowledge retention (88.46%) and faithfulness (82.54%), though its relevancy score is slightly lower (84.02%) than Mistral2-7b (vanilla)" [9].

One challenge with fine tuning is how expensive it is as it takes a lot of computing power. There are ways to reduce the computing power including parameter efficient finetuning, quantization, and distributed training. Despite this, fine tuning can be really expensive. The model can also experience something called catastrophic-forgetting during fine-tuning. Catastrophic forgetting occurs when specializing too much on one dataset can lead it to forget general knowledge.

To circumvent the massive compute requirements of full fine-tuning, Parameter Efficient Fine-Tuning methods have been developed. Low-Rank Adaptation (LoRA) is a commonly used fine tuning technique that freezes pre-trained model weights and injects trainable rank decomposition matrices into the frozen model architecture [10]. This method reduces the number of trainable parameters, making it feasible to fine-tune large models on consumer-grade hardware.

While the efficacy of LoRA is well-documented on large

datasets, there is a gap in the literature regarding its effectiveness in ultra-low-resource medical environments. Our project aims to bridge this gap by evaluating whether LoRA can successfully align a model’s reasoning structure using a highly constrained dataset from PubMedQA [11]. Furthermore, we investigate how different baseline architectures interact with Zero-Shot, Few-Shot, and Chain-of-Thought prompting strategies both before and after fine-tuning. By doing so, we seek to determine the minimum data threshold required to mitigate systemic clinical bias and improve reasoning coherence in lightweight LLMs.

II. METHODS

The goal of our project was to determine if we can mitigate bias in medical question and answering using LoRA [12] to adapt a model’s reasoning structure using a low-resource dataset. To determine if this goal was feasible, we used the following methodology.

A. Dataset and Preprocessing

To evaluate our models on their medical questioning and answering knowledge, we utilized the PubMedQA dataset [11], a biomedical question and answering dataset designed to answer clinical questions with “yes”, “no”, or “maybe”, alongside a long-form explanatory rationale. To ensure sound evaluation and avoid degradation over cross-domain knowledge, we isolated the task environment by using a labeled subset of the dataset. We performed an 80/20 train-test split on the full corpus of this dataset, and from this baseline, we simulated an ultra-low-resource environment to explicitly test the data efficiency of our fine-tuning methods.

We extracted 20 samples from the 80% training partition to serve as our fine-tuning dataset, and selected 20 constant, hold out examples from the 20% testing partition for final evaluation. This sampling design ensures our models were evaluated on unseen clinical scenarios, allowing us to accurately measure LoRA’s ability to achieve structural alignment and Chain-of-Thought adaptation on a highly constrained data budget.

B. Model Selection and Baselines

To evaluate systemic biases and domain struggles, we established a multi-tiered baseline strategy comparing general-purpose foundation models against a medical-domain-specific model. For our medical-domain-specific model, we used the BioMistral [13] model, an open-source LLM utilizing Mistral-7b as the backbone and pre-trained on data from PubMed Central. For our general foundation models, we used the LLaMA-3-8B [14] and Gemma-7B [15] models as they were commonly used open source models on hugging face that we had access to.

C. Prompt Engineering Interventions

We evaluated all base models across three distinct zero-shot and in-context learning strategies to measure how formatting impacts clinical hallucination and coherence. To do so, we used Zero-Shot, Few-Shot, and Chain-of-Thought prompting.

1) **Zero-Shot:** The model was provided the PubMed abstract context and the question, and asked to output a direct answer. An example can be seen down below:

- “Question: Sternal skin conductance: a reasonable surrogate for hot flash measurement? Answer:”

2) **Few-Shot:** The model was provided with a static, pre-solved, high-quality example from the training set within the system prompt to establish a structural pattern before answering the target question. For fewshot, we used the example answers from the dataset “long-answer” so that we didn’t have to run it on multiple questions at a time. Note this may skew the performance of the model since it is inherently given more context than the other methods; however, it is still an environment where we can measure the effectiveness of LoRA. An example can be seen down below:

- “Question: Is eligibility for a chemotherapy protocol a good prognostic factor for invasive bladder cancer after radical cystectomy?

Answer: These data suggest that being willing and fit enough for a chemotherapy protocol is a good prognostic factor for invasive bladder cancer. This eligibility bias emphasizes the need for prospective... [shortened for brevity]

Question: Delayed imaging in routine CT examinations of the abdomen and pelvis: is it worth the additional cost of radiation and time?

Answer: An additional delayed phase through the abdomen in routine contrast-enhanced CT examinations of the abdomen and ... [shortened for brevity]

Question: Does the SCL 90-R obsessive-compulsive dimension identify cognitive impairments? Answer: Our data suggest that the SCL 90-R is best viewed as an indicator of unidimensional [shortened for brevity]

Question: Sternal skin conductance: a reasonable surrogate for hot flash measurement? Answer:”

3) **Chain-of-Thought:** The model was explicitly instructed to “think step-by-step,” generating a detailed reasoning path before outputting its final classification. An example can be seen down below:

- “Question: Sternal skin conductance: a reasonable surrogate for hot flash measurement? Let’s think step by step before giving a final answer. Reasoning:”

D. Parameter Efficient Fine Tuning

To address the hardware constraints of training 7B and 8B parameter models, we utilized a Quantized Low-Rank Adaptation (QLoRA) [12] pipeline. All fine-tuning, inference generation, and evaluation were executed using Google Colab Pro instances provisioned with Nvidia T4 GPUs. The base models were loaded in 4-bit precision using BitsAndBytesConfig (NF4 quantization type with double quantization) to reduce VRAM overhead. For our LoRA Configuration, we targeted the q_{proj} and v_{proj} attention blocks using a rank of 16, an alpha of 32, and a dropout rate of 0.05. The models were fine-tuned using the SFTTrainer from Hugging Face trl library, formatting

the training split into prompt-response pairs that mirrored our Chain-of-Thought reasoning structure.

E. Evaluation Framework

To evaluate the performance of our model on the testing dataset, we used two main methods to judge our models' correctness: Semantic Accuracy and Clinical Coherence. To do this step, we separated the generation step from our evaluation step. Our model's outputs were saved and scored by the DeepEval [16] framework. For semantic accuracy, we calculated the cosine similarity between the model's generated rationale and the ground truth LONG-ANSWER to measure factual alignment. We decided to use a lightweight sentence transformer "all-MiniLM-L6-v2" [17] to speed up the embedding process. While we could have used a sentence transformer specific to BioMed QA, we were faced with resource and memory constraints. To measure Clinical Coherence, we implemented G-Eval [16], an LLM-as-a-judge metric to score the quality of reasoning. We used the LLaMA-3.1-8b-instant model as our scoring LLM. To ensure the reliability of this model, we manually validated a random sub-sample of the judge's scores against human expert intuition to confirm it correctly adhered to our G-Eval rubric without hallucinating grading criteria. We defined a custom rubric that penalized hallucinated medical facts and rewarded logical synthesis of the provided abstract. This allowed us to mathematically track shifts in systemic biases and coherence across different prompting methods and medical domains.

III. EXPERIMENTS AND RESULTS

To accurately measure the impact of our low resource fine-tuning, we evaluated the models against their respective out-of-the-box performance. Our baselines consisted of the pre-trained, unmodified versions of BioMistral [13], Gemma-7B [15], and LLaMA-3-8B [14]. The experimental variables were the prompting strategy of Zero-Shot (ZS) vs Few-Shot (FS) vs Chain-of-Thought (CoT), and the model state of the base unmodified model vs the LoRA fine-tuned model.

As mentioned in the methods section, to quantitatively evaluate the model outputs, we compared factual alignment to the dataset using cosine similarity, and to measure reasoning quality and bias mitigation, we used a custom LLM-as-a-judge metric to score Clinical Coherence of the rationales.

A. Discussion

Table I summarizes the average Cosine Similarity and Clinical Coherence scores across all 20 test samples for the base and LoRA-adapted models under different prompting strategies. Figure 1 shows an overview of the differences between the base and LoRA fine-tuned models for both Cosine Similarity and Clinical Coherence. We found that across all models for zero-shot prompting, using our low resource fine-tuning resulted in significant performance gains in cosine similarity. The base versions of BioMistral and Gemma scored 0.280 and 0.320 respectively on Cosine Similarity using zero-shot prompting, and upon fine-tuning their performance increased

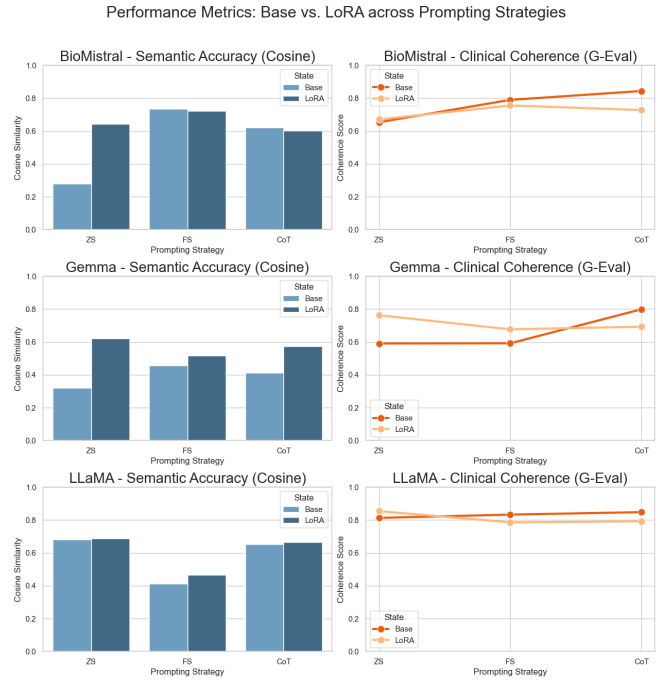


Fig. 1. Right: Bar graphs showing the Cosine Similarity scores for all three models across all three different prompting types. Light blue is the base model, dark blue is the LoRA fine-tuned model. Left: Line graphs showing the Clinical Coherence scores for all three models across all three different prompting types. Red is the base model, orange is the LoRA fine-tuned model.

to 0.645 and 0.620 respectively, even with using only 20 samples for fine-tuning. The LLaMA model also saw performance gains, though not as drastic compared to BioMistral and Gemma for zero-shot prompting. All models saw improvement in cosine similarity across all prompting strategies except for BioMistral being prompted using the few-shot and chain-of-thought prompting strategies. To highlight this qualitative leap, consider how BioMistral approached the test question: 'Does early discharge with nurse home visits affect the adequacy of newborn metabolic screening?' The base zero-shot model outputted 'No' with no other explanatory rationale, whereas the LoRA-tuned zero-shot model successfully adopted the target structure, outputting: 'No, early discharge with nurse home visits does not affect the adequacy of newborn metabolic screening. Context: Early discharge of newborns from the hospital is a common practice...'

For clinical coherence, across zero-shot prompting, all three models saw marginal performance gains from base to fine-tuned, the Gemma model seeing the greatest increase in performance from 0.591 to 0.763. However, for the other two prompting strategies, other than with Gemma for few-shot prompting, there were performance decreases from the base model to fine-tuning. This result might have occurred due to the model being fed too much context in the few-shot prompting that might not have necessarily related to the abstract the model was supposed to gain context from, causing the model to get "distracted" in some sense from being exposed to irrelevant context.

Overall, our method improved performance in cosine simi-

TABLE I
MEAN SEMANTIC ACCURACY (COSINE SIMILARITY) AND CLINICAL COHERENCE RESULTS

Model	ZS Cosine	ZS Coh.	FS Cosine	FS Coh.	CoT Cosine	CoT Coh.
BioMistral Base	0.280	0.654	0.735	0.790	0.620	0.844
BioMistral LoRA	0.645	0.671	0.723	0.756	0.601	0.728
Gemma Base	0.320	0.591	0.458	0.592	0.414	0.799
Gemma LoRA	0.620	0.763	0.516	0.677	0.573	0.693
LLaMA Base	0.680	0.813	0.412	0.833	0.654	0.848
LLaMA LoRA	0.686	0.854	0.466	0.786	0.664	0.792

larity for our generalized models across all prompting types, and led to performance improvements in zero-shot prompting for Clinical Coherence.

B. Limitations

Since our fine-tuning intervention was restricted to only a 20-sample training set, the performance gains observed across our models were primarily due to the models being able to better represent the formatting and the alignment of the answers rather than acquiring new biomedical knowledge, especially for the zero-shot prompted models. The models learned primarily how to parse the abstract context and structure logical rationales based on their pre-existing weights.

Another limitation was our LLM-as-a-judge model, the LLaMA-3.1-8B model. Although this judge model is good for comparison tasks and has more robust alignment and instruction following capabilities compared to the other three models we tested, and while we mitigated grading hallucinations by manually validating a sub-sample of the judge’s scores, using a more powerful frontier model such as the 405 billion parameter LLaMA-3.1 model or the frontier GPT-5 model or another very large model would make us feel more confident about our clinical coherence metric score. Using a model with a large context window as the judge may have scored the models we tested differently.

In the future, we plan on experimenting with tuning how many samples we present in the training set to see if there is a point where the models learn how to structure their answers better and acquire new biomedical knowledge. A future experiment could also include not only using a stronger LLM-as-a-judge model for measuring Clinical Coherence, but also using multiple judges to score this criterion better.

IV. CONCLUSION

Mitigating LLM Bias using fine-tuning, especially with a small dataset is a challenging task. In our project, we proposed a framework using LoRA to fine-tune both medical LLMs and generalized LLMs on a small dataset across multiple prompting strategies such as zero-shot, few-shot, and Chain-of-Thought prompting. We have shown that by applying our framework, we are able to see significant increases in cosine similarity using a zero-shot prompting method and similar smaller increases across other prompting methods, indicating

that despite using a small dataset, we were able to still get increases in performance with our framework.

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